



Barcelona School of Economics

Assignment 6

Econometrics

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1 Randomized Control Trials

1.1 Baseline and Experiment Integrity

Table 1: Baseline Data: Teacher and Student Attendance

	Treatment	Control	Difference (T-C)
Teacher attendance			
School open (mean attendance)	0.659 (0.480)	0.641 (0.486)	0.018
Observations	41	39	80
Student participation			
Number of students present	11.659 (11.051)	10.205 (9.937)	1.454
Observations	41	39	80

To assess whether the randomization was successful, we compare baseline (time = 1) teacher attendance and student attendance across treatment and control schools. Table 1 shows that mean teacher attendance is 0.641 in the control group and 0.659 in the treatment group. Student attendance averages 10.205 students in control schools and 11.659 students in treatment schools. The differences in both variables are small relative to their standard deviations, and there is no indication of systematic pre-treatment differences between groups. This suggests that the randomization generated comparable treatment and control schools at baseline, supporting the validity of subsequent treatment effect estimates.

1.2 Results

Table 2: After program start: Average Teacher Attendance

	September 2003 – February 2006		
	Treatment	Control	Diff (T-C)
Teacher attendance	0.787 (0.030)	0.580 (0.023)	0.207*** (0.030)
Observations	1,575	1,496	3,071

We estimate the causal effect of the teacher monitoring program using post-treatment observations (*time* > 1). Average teacher attendance in treatment schools was 0.787, compared to 0.580 in control schools. The estimated difference, 0.207, is statistically significant at the 1% level (clustered standard error 0.030). This implies that the Seva Mandir program increased teacher attendance by roughly 21 percentage points relative to the control group

and thus achieving its intended goal. This is consistent with the findings reported in Duflo, Hanna, and Ryan (2012).

2 Matching

2.1 Propensity Score Index

Table 3 shows that conditional independence is not violated as the balancing property is satisfied across all the blocks for each different treatment (displaced, separation, and mass layoff) generated by the algorithm. This shows propensity score stratification to be adequate in precedence to matching.

Table 3: Summary of Blocking Structure After Stratification

Statistic	Displaced	Separation	Mass Layoff
Final number of blocks	6	6	4
Balancing property satisfied	Yes	Yes	Yes
Smallest block size (N)	25	25	4,667
Largest block size (N)	6,108	7,671	9,717
Treated units total	3,396	3,033	363
Controls total	17,598	17,961	20,545

However, there are some notes on the reliability and robustness of the results:

- Table 4 shows a great imbalance cross treatments (workers displaced, separated, or laid off in 1999). This heavily exhibits the need for propensity score matching in order to construct fair comparisons.
- The pseudo- R^2 (0.0028) of the logistic regression on the treatment is low, as seen in Table 5. This means that regardless of differences in covariates, the observations are likely to have lower variance in propensity, as seen in the narrow region of common support [0.1172, 0.2209].
- Given the small region of common support in Table 6, the resulting blocks ended up with some being empty (block 1) or with very few observations (block 6). This makes comparisons for these blocks possibly unstable and likely to have residual confounding.

Table 4: Treatment Prevalence in 1998

Treatment	Controls	Treated	Total
Displaced	17,598	3,396	20,994
Separation	17,961	3,033	20,994
Mass Layoff	20,631	363	20,994

Table 5: Summary of Logistic Regression Models for Propensity Score Estimation

	Displaced	Separation	Mass Layoff
female	—***	—***	+**
earnings_1997	—**	n.s.	—*
decade_brth	+***	+***	n.s.
Pseudo R^2	0.0028	0.0037	0.0046
Observations	20,994	20,994	20,994

Notes: Coefficients shown by sign only. Significance: * p_i0.10, ** p_i0.05, *** p_i0.01.

Table 6: Common Support Summary Across Treatments

Statistic	Displaced	Separation	Mass Layoff
Common support range	[0.117, 0.221]	[0.100, 0.204]	[0.0128, 0.0248]
Mean propensity score	0.162	0.144	0.0173
Std. deviation	0.0182	0.0193	0.00375
Skewness	0.164	0.094	0.603
Kurtosis	2.49	2.39	2.01

Regardless, we may proceed with matching.

2.2 Nearest Neighbor and Kernel Matching

Following the estimation of ATT by CP, we obtain the estimates for ATT and the standard errors for both nearest neighbor matching and kernel (with bootstrapped standard errors with 10 iterations) matching. Table 7 shows the results.

Table 7: ATT Estimates (Log Earnings) using Nearest Neighbor and Kernel Matching

Panel A: Displaced Workers				
Year	NN ATT	NN SE	Kernel ATT	Kernel SE
1995	0.002	0.010	-0.016	0.012
1996	-0.007	0.007	-0.015	0.007
1997	-0.009	0.007	-0.016	0.006
1998	-0.023	0.007	-0.026	0.007
1999	-0.077	0.009	-0.077	0.006
2000	-0.103	0.010	-0.102	0.007
2001	-0.040	0.010	-0.037	0.010
Panel B: Separated Workers				
Year	NN ATT	NN SE	Kernel ATT	Kernel SE
1995	0.000	0.011	-0.011	0.010
1996	-0.002	0.007	-0.009	0.007
1997	-0.004	0.007	-0.008	0.009
1998	-0.022	0.007	-0.021	0.006
1999	-0.070	0.009	-0.067	0.010
2000	-0.099	0.010	-0.093	0.011
2001	-0.036	0.011	-0.029	0.007
Panel C: Workers in Mass Layoffs				
Year	NN ATT	NN SE	Kernel ATT	Kernel SE
1995	0.011	0.029	-0.053	0.025
1996	-0.037	0.022	-0.071	0.014
1997	-0.043	0.023	-0.076	0.020
1998	-0.030	0.022	-0.066	0.014
1999	-0.100	0.028	-0.139	0.033
2000	-0.107	0.030	-0.152	0.036
2001	-0.053	0.031	-0.103	0.035

Both estimators show the same overall pattern: a sharp earnings drop in 1999 followed by only partial recovery. Pre-treatment differences from 1995–1998 are minimal, the treatment year shows a clear negative shock, and the post-1999 path reflects a slow recovery that never returns to baseline. The fact that both NN and kernel matching produce this same shape strengthens confidence that the results are not driven by the choice of estimator.

Kernel matching generally yields larger earnings losses, especially for displacement, because it uses a wider set of comparison observations and therefore uncovers more of the downward shift in counterfactual earnings. In contrast, NN matching may exclude slightly poorer matches and, in areas of tight common support, may repeatedly reuse the same controls, resulting in noisier and often smaller effects. Kernel matching, by borrowing strength from many nearby controls, produces smoother and often more negative estimates.

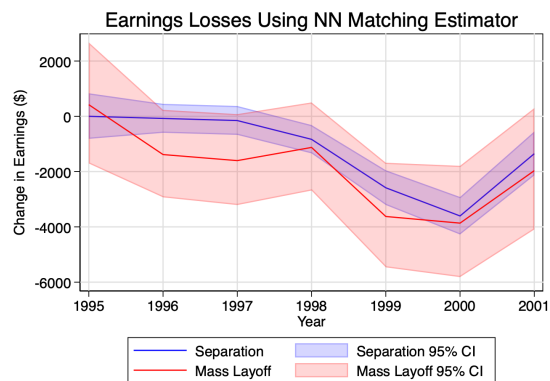
We also see that NN estimates fluctuate more from year to year, while kernel estimates are smoother with lower standard errors, particularly for separations and mass layoffs. This follows from kernel averaging over many controls, which reduces variance but can introduce some bias when support is thin. This smoothing helps explain why kernel estimates show more persistent negative post-treatment effects.

2.3 Presenting Results

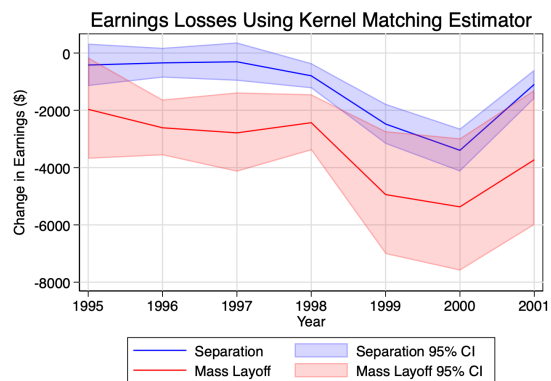
VWH estimates of earnings losses from matching are shown in Figure 1. Comparing these to estimates from the JLS model on the VWH data, the overall patterns are consistent, with some differences:

- Earnings losses remain negative (or close to zero in the first period), in line with the effects of job displacement estimated by JLS and CP.
- The sharp drop in 1999 is captured by both estimators. The kernel estimator better distinguishes the shock differences between workers separated and those affected by mass layoffs, and this gap persists during the slow post-treatment recovery.
- Recovery is observed across NN, kernel, and JLS estimates. However, NN and kernel estimates show a persistent drop in earnings immediately following the shock (2000) before recovering, whereas JLS estimates indicate an immediate slow recovery for separated workers, with mass-layoff workers recovering a year later.

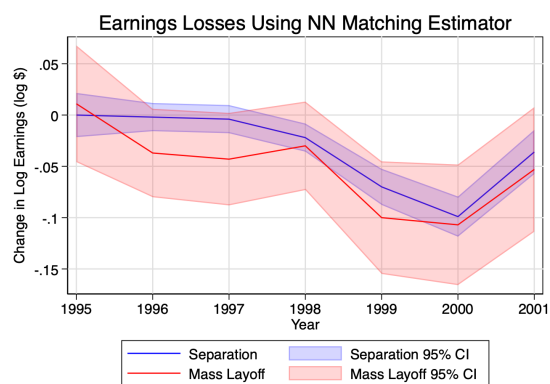
Figure 2 aggregates the effects of all displaced workers. The general trend mirrors the sharp negative shock in 1999, followed by recovery. Kernel estimates capture pre-shock stability and post-shock recovery more smoothly than NN, though both methods clearly show the immediate negative impact of job displacement and the subsequent movement toward recovery.



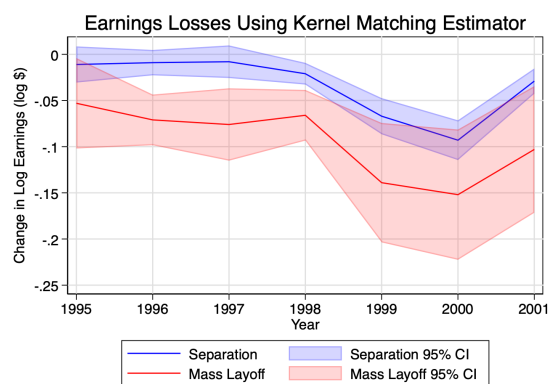
(a) Nearest Neighbors, Real Earnings



(b) Kernel, Real Earnings

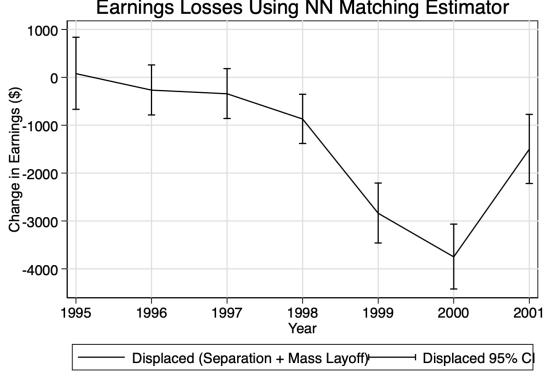


(c) Nearest Neighbors, Log Earnings

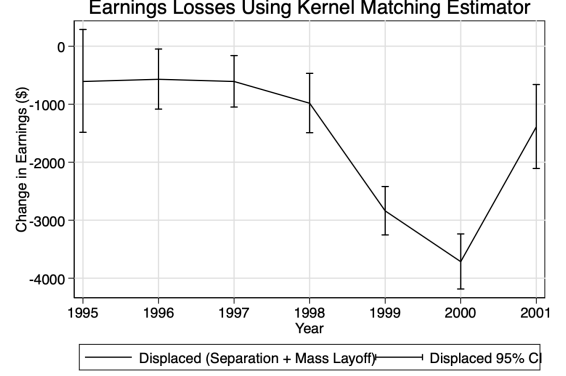


(d) Kernel, Log Earnings

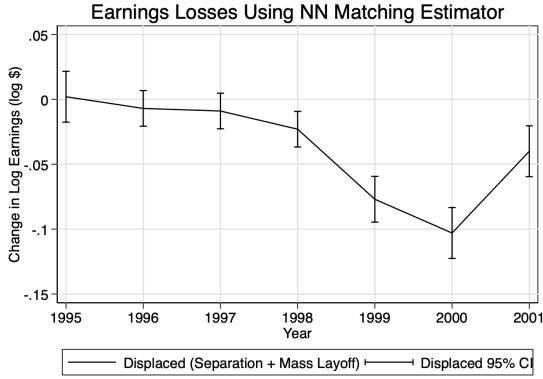
Figure 1: Estimated Earnings Losses (USD) from CP ATT Estimation for Workers Separated and Part of Mass Layoffs



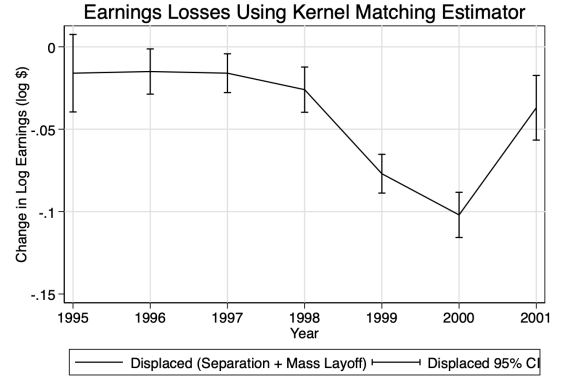
(a) Nearest Neighbors, Real Earnings



(b) Kernel, Real Earnings



(c) Nearest Neighbors, Log Earnings



(d) Kernel, Log Earnings

Figure 2: Estimated Earnings Losses (USD) from CP ATT Estimation for Workers Displaced (Separated and Mass Layoffs)

3 Instrumental Variables

3.1 OLS and Probit Estimates

We study the effect of fertility on mothers' labor supply using the following empirical model:

$$workedm_i = \beta_0 + \beta_1 kidcount_i + X_i' \beta + \varepsilon_i, \quad (1)$$

where $workedm_i$ is a binary indicator equal to 1 if the mother is employed, $kidcount_i$ is the number of children, and X_i is a vector of control variables (mother's age and race dummies). We estimate the model using Ordinary Least Squares (OLS) and a probit specification.

OLS Estimates

Table 8 reports the linear probability model estimates.

Table 8: OLS (Linear Probability Model) Estimates

Variable	Coefficient	Std. Error	t-statistic	p-value
kidcount	-0.09107	0.00096	-94.67	0.000
agem	0.01469	0.00022	66.36	0.000
blackm	0.15067	0.00239	63.12	0.000
hisp	-0.00831	0.00452	-1.84	0.066
othracem	0.02751	0.00463	5.94	0.000
Constant	0.33766	0.00686	49.21	0.000
Observations	400,169			
R-squared	0.0344			

Probit Estimates and Marginal Effects

We next estimate a probit model for employment. Table 9 shows the probit coefficients, and Table 10 presents average marginal effects.

Table 9: Probit Model Estimates

Variable	Coefficient	Std. Error	z-statistic	p-value
kidcount	-0.23707	0.00255	-92.99	0.000
agem	0.03826	0.00058	66.03	0.000
blackm	0.40394	0.00642	62.90	0.000
hisp	-0.02034	0.01177	-1.73	0.084
othracem	0.07178	0.01209	5.94	0.000
Constant	-0.42635	0.01787	-23.86	0.000
Observations	400,169			
Pseudo R-squared	0.0256			

Table 10: Average Marginal Effects from Probit Model

Variable	AME	Std. Error	z-statistic	p-value
kidcount	-0.09072	0.00094	-96.20	0.000
agem	0.01464	0.00022	67.16	0.000
blackm	0.15458	0.00242	63.86	0.000
hisp	-0.00778	0.00450	-1.73	0.084
othracem	0.02747	0.00463	5.94	0.000

While the OLS and probit models can be estimated and interpreted as correlations, they do not yield a causal estimate of the effect of fertility on mothers' employment. The key econometric issue is **endogeneity of fertility decisions**. The variable *kidcount* is likely correlated with unobserved determinants of labor supply such as:

- preferences for family versus career,
- socioeconomic background,
- labor market attachment,
- unobserved ability or motivation.

These factors directly influence both fertility and employment, violating the exogeneity condition

$$\mathbb{E}[\varepsilon_i \mid \textit{kidcount}_i] = 0.$$

Thus, both OLS and probit estimates are biased and inconsistent due to omitted-variable bias. The nonlinear probit specification does not resolve this issue, as the endogeneity problem arises from the correlation between regressors and unobservables, not from the functional form.

The OLS and probit models estimated above allow descriptive analysis but are **not valid methods** for estimating the causal impact of additional children on mothers' labor supply. Instrumental variables are required to address the endogeneity of fertility decisions.

3.2 IV Probit Estimation

We estimate the structural model:

$$\textit{workedm}_i = \beta_0 + \beta_1 \textit{kidcount}_i + X_i' \beta + \varepsilon_i,$$

where *workedm_i* is a dummy equal to 1 if mother *i* is employed, *kidcount_i* is the total number of children, and *X_i* contains controls for age and race. Since fertility may be endogenous, we instrument *kidcount_i* using *twin_latest*, a dummy equal to 1 if the most recent birth was a twin birth.

The IV-probit specification is:

$$P(\textit{workedm}_i = 1) = \Phi(\gamma_0 + \gamma_1 \widehat{\textit{kidcount}}_i + X_i' \gamma + u_i),$$

where $\widehat{\textit{kidcount}}_i$ is predicted from the first stage.

First-stage regression

We estimate the linear first-stage regression of fertility on the instrument and controls:

Table 11: First-stage regression: *kidcount* on *twin_latest* and controls

Variable	Coefficient	Std. Error	t	p-value
twin_latest	0.3851	0.00995	38.72	0.000
agem	0.03097	0.00036	86.07	0.000
blackm	0.32368	0.00388	83.40	0.000
hispm	0.43705	0.00739	59.17	0.000
othracem	0.12093	0.00759	15.93	0.000
Constant	1.56457	0.01098	142.50	0.000
Observations	400,169			
R-squared	0.0404			

Interpretation. The coefficient on *twin_latest* is large and highly significant ($t = 38.72$), showing that twin births substantially and exogenously increase the number of children. Thus, the relevance condition for a valid instrument is clearly satisfied.

IV-Probit results

Table 12 reports the IV-probit coefficient estimates (latent index).

Table 12: IV-Probit estimates (latent index)

Variable	Coefficient	Std. Error	z	p-value
kidcount (instrumented)	-0.07159	0.04132	-1.73	0.083
agem	0.03289	0.00152	21.66	0.000
blackm	0.34737	0.01610	21.57	0.000
hispm	-0.09154	0.02108	-4.34	0.000
othracem	0.05154	0.01313	3.93	0.000
Constant	-0.68005	0.06405	-10.62	0.000

Additional statistics:

- $\text{corr}(e_{\text{kidcount}}, e_{\text{workedm}}) = -0.1311$ ($p = 0.0001$)
- Wald test of exogeneity: $\chi^2(1) = 16.12$, $p = 0.0001$

These results reject the null of exogeneity, confirming that standard probit or OLS would be inconsistent and that IV methods are necessary.

Average Marginal Effect

Using `margins, dydx(kidcount) predict(pr)`, we compute the average marginal effect (AME) on the probability scale:

Table 13: Average marginal effect (IV-Probit)

Variable	AME	Std. Error
kidcount	-0.02763	0.01583

Interpretation. For mothers whose fertility is affected by having twins (the compliers), one additional child reduces the probability of being employed by about **2.8 percentage points**. The effect is smaller and less precisely estimated than in the OLS or standard probit model, consistent with the IV estimate capturing the LATE for a specific subgroup.

Critique of the `twin_latest` instrument

Relevance (Strength). Twin births strongly increase fertility, as shown by the large coefficient (0.385) and t -statistic (38.72). The instrument is therefore highly relevant.

Exclusion restriction. This assumption is more controversial. The instrument requires that twin births affect maternal employment only through the number of children. Possible violations include:

- Twins likely impose a much larger immediate childcare burden, directly reducing employment beyond the effect of “just one more child.”
- Twin pregnancies are medically riskier and may affect the mother’s long-term health or physical ability to work.
- The probability of twins increases with maternal age or fertility treatments, potentially correlating with labor market characteristics.

If any of these mechanisms influence employment directly, the exclusion restriction fails.

Local Average Treatment Effect (LATE). The IV estimate identifies the effect of an additional child *only* for mothers whose fertility increased because they happened to have twins. This group may not be representative of all mothers, so external validity is limited.

Conclusion

The IV-probit results indicate that additional children reduce employment probabilities, with an estimated effect of roughly 2.8 percentage points for compliers. The instrument is strong and plausibly exogenous, although exclusion restriction concerns remain. The IV estimates are smaller and less precise than the naïve estimates, consistent with correction for positive endogeneity bias.

3.3 Marginal Effects

We compute and plot the marginal effect of an additional child on the probability that a mother is employed, using the IV–probit model estimated in the previous question (instrument: `twin_latest`). Because probit coefficients are defined on a latent index, we report marginal effects on the probability scale. These are obtained in Stata with `margins, dydx(kidcount) predict(pr)`.

Table 14: Average marginal effect of an additional child (IV–Probit)

Variable	AME (Pr)	Std. Error
<code>kidcount</code>	-0.02763	0.01583

Interpretation. The IV–probit average marginal effect implies that, for mothers whose fertility is increased by a twin birth (the compliers), one additional child reduces the probability of employment by approximately **2.8 percentage points** (AME = 0.0276).

Marginal effects by number of children and plot

We also compute marginal effects evaluated at integer values of `kidcount` and plot them to show how the effect varies across family sizes. In this dataset the marginal effect is nearly constant across the observed range of children, so the plotted line is approximately flat and centered near the AME reported above. The table below illustrates the output of Stata’s `margins, dydx(kidcount) at(kidcount=(1(1)6)) predict(pr)`.

Table 15: Marginal Effect of an Additional Child at Different Family Sizes

Kidcount	AME	Std. Error	95% CI
1	-0.02717	0.01518	[-0.05693, 0.00259]
2	-0.02764	0.01580	[-0.05860, 0.00332]
3	-0.02797	0.01618	[-0.05969, 0.00375]
4	-0.02817	0.01633	[-0.06018, 0.00385]
5	-0.02822	0.01624	[-0.06005, 0.00361]
6	-0.02814	0.01590	[-0.05930, 0.00303]

Interpretation

The marginal effect of an additional child is consistently negative across all family sizes. Although the estimates vary slightly (from -0.0272 to -0.0282), the differences are economically negligible. This occurs because `kidcount` enters the IV–probit specification linearly, so changes in the marginal effect across family sizes arise only through small changes in the normal density term $\phi(X\hat{\beta})$.

Thus, the marginal effect curve is *nearly flat*, indicating that:

Each additional child reduces the probability of maternal employment by approximately 2.7–2.8 percentage points, and this effect is effectively constant across different numbers of children.

The corresponding `marginsplot` shows a nearly horizontal line, reflecting the stability of the marginal effect across values of `kidcount`.

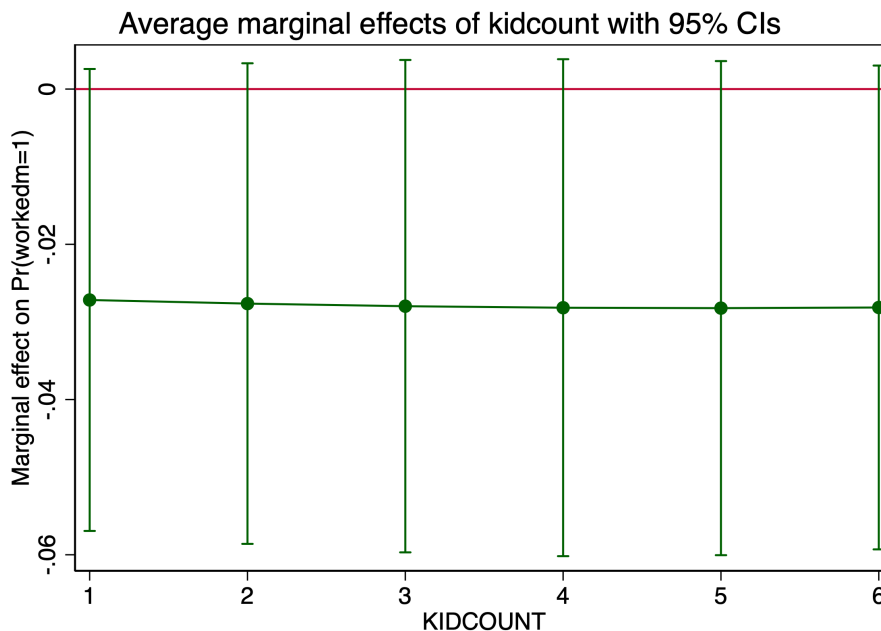


Figure 3: Marginal effect of an additional child on the probability of maternal employment (95% CI).

The near-flat shape of the plot is consistent with a roughly constant marginal effect across values of `kidcount` in this sample and model. It means the effect per additional child is similar whether the family has 1, 2, 3, or more children.